Mathematical and Computational Methods

# Methods

## Part III: Differentiation (Units 7–8)

Most problems on the exercise sheets are instances of a few standard types, and each type has a *method*: a fixed sequence of steps, taken from the notes, that solves every problem of that type. This guide collects them, unit by unit. Each method states the problem it solves, lists its steps and works one example that is not on any sheet. It ends with the mistakes to avoid, where the notes teach it, and a link to a Review set of problems of that kind. The units fall into four parts: foundations and complex numbers, vectors and matrices, differentiation, and integration.

The guide does not say which sheet problem needs which method. Recognising the type is part of what the sheets train, and in an exam nothing tells you. Where a unit's problems look alike but need different methods, its chapter opens with a table for choosing.

## Contents


**Part III Differentiation** 2

**Unit 7 Differentiation I: limits and the rules** . . . 2
- 7.1 Evaluating a limit . . . 2
- 7.2 Continuity and differentiability at a join . . . 4
- 7.3 Differentiating from first principles . . . 5
- 7.4 Choosing between the product, quotient and chain rules . . . 6
- 7.5 Implicit differentiation . . . 7
- 7.6 Parametric differentiation . . . 8
- 7.7 Logarithmic differentiation . . . 9

**Unit 8 Differentiation II: applications** . . . 11
- 8.1 Tangent and normal . . . 11
- 8.2 A root exists and is unique (IVT and monotonicity); the mean value theorem 12
- 8.3 l'Hôpital's rule . . . 13
- 8.4 Classifying critical points; the closed-interval method . . . 15
- 8.5 Sketching a curve by the checklist, or from a graph of $f'$ . . . 16
- 8.6 An optimisation word problem . . . 17
- 8.7 Newton's method and gradient descent . . . 18

# Part III Differentiation

Units 7–8: limits, the rules of differentiation, and what a derivative is for, from tangents and the existence of roots to extreme values, sketches, optimisation and iterative methods. Most problems start by finding a derivative; the question decides what to do with it.

## Unit 7 Differentiation I: limits and the rules

Seven methods: two for limits, one that finds a derivative as a limit, and four that differentiate. Which of the four depends on how the function is given. A tangent line, a velocity, a higher derivative and a sensitivity are each one step after the rules, and are variants of Method 7.4.

### Which method?

| What you see | What to do |
|---|---|
| one formula on both sides of $a$, and substituting gives a number | 7.1: that number is the limit |
| $\frac{0}{0}$, and top and bottom share a factor | 7.1: factor and cancel |
| $\frac{0}{0}$ with a square root, or with $1 - \cos x$ | 7.1: multiply by the conjugate |
| $\frac{0}{0}$ with the sine of something small | 7.1: rewrite to $\frac{\sin u}{u}$ |
| (a nonzero number)/0 | 7.1: one-sided limits; an asymptote |
| $x \to \infty$ with a rational function, or a difference with a root | 7.1: conjugate first for a root; divide by the highest power |
| a join in a piecewise function | 7.2: one-sided limits at the join |
| "from first principles" | 7.3: the difference quotient |
| a product or quotient of two factors, a function of a function | 7.4: name $u$ and $v$, or outer and inner |
| $y$ tangled with $x$ in an equation | 7.5: implicit differentiation |
| $x$ and $y$ both given in terms of $t$ | 7.6: parametric differentiation |
| the variable in both a base and an exponent; a product or quotient of three or more factors | 7.7: take logarithms first |

### 7.1 Evaluating a limit

**Problem.** Find $\lim_{x\to a} f(x)$, the one-sided limits at $a$, or $\lim_{x\to\infty} f(x)$, and the asymptotes they give.

**Method**

1. If $f$ is one formula on both sides of $a$, substitute $x = a$ into it. If the result is a number, that is the limit. At the join of a piecewise function, find the two one-sided limits instead (Method 7.2, step 2).
2. If it is $\frac{0}{0}$, simplify for $x \neq a$: factor and cancel the common factor; multiply top and

bottom by the conjugate of a root (for $1-\cos x$, by $1+\cos x$, then $1-\cos^2 x=\sin^2 x$); or rewrite to $\dfrac{\sin u}{u}$ with $u\to 0$, using $\lim_{u\to 0}\dfrac{\sin u}{u}=1$.

3. Substitute into the simplified form, and state the limit.
4. If it is (a nonzero number)/0, find the limits from below and from above, $x\to a^-$ and $x\to a^+$, from the signs of the top and the bottom. An infinite one gives the vertical asymptote $x=a$. The limit exists only when the two one-sided limits exist and agree.
5. For $x\to\infty$, divide the top and the bottom by the highest power of $x$ in the denominator. Each $1/x^k$ tends to 0: write the $\dfrac{(\cdot)-0}{(\cdot)+0}$ line. For a difference with a root, multiply and divide by its conjugate first, as in step 2.
6. If the limit is a number $L$, state it and the horizontal asymptote $y=L$. For a rational function the same $L$ holds as $x\to-\infty$; for any other function, check that direction separately. If a power of $x$ is left on the top, the limit is infinite and there is no horizontal asymptote.

**Example.** (a) For $f(x)=\dfrac{2x^2-5x+3}{x^2-x}$, find the limits of $f$ as $x\to 1$, as $x\to 0^{\pm}$ and as $x\to\infty$, and the asymptotes. (b) Find $\lim\limits_{x\to 2}\dfrac{\sqrt{2x+5}-3}{x^2-4}$. (c) Find $\lim\limits_{x\to 0}\dfrac{\sin 3x}{x^2+2x}$.

(a) At $x=1$ substitution gives $\frac{0}{0}$. Factor and cancel, for $x\neq 1$:

$$f(x)=\frac{(2x-3)(x-1)}{x(x-1)}=\frac{2x-3}{x}\longrightarrow\frac{2-3}{1},$$

so $\boxed{\lim\limits_{x\to 1}f(x)=-1}$. At $x=0$ substitution gives $\frac{3}{0}$. Near 0 use the simplified form: its top tends to $-3$ and its bottom $x$ has the sign of $x$, so $f\to-\infty$ as $x\to 0^+$ and $f\to+\infty$ as $x\to 0^-$. The vertical asymptote is $\boxed{x=0}$. As $x\to\infty$, divide by $x^2$:

$$f(x)=\frac{2-\frac{5}{x}+\frac{3}{x^2}}{1-\frac{1}{x}}\longrightarrow\frac{2-0+0}{1-0}=2.$$

The horizontal asymptote is $\boxed{y=2}$, approached as $x\to-\infty$ too.

(b) Substitution gives $\frac{0}{0}$. Multiply top and bottom by $\sqrt{2x+5}+3$; the top becomes $(2x+5)-9=2(x-2)$, and the bottom factors as $(x-2)(x+2)(\sqrt{2x+5}+3)$. Cancel $x-2$:

$$\frac{\sqrt{2x+5}-3}{x^2-4}=\frac{2}{(x+2)(\sqrt{2x+5}+3)}\longrightarrow\frac{2}{4\cdot 6}=\boxed{\frac{1}{12}}.$$

(c) Substitution gives $\frac{0}{0}$. Factor the bottom as $x(x+2)$ and supply the 3 that $\frac{\sin u}{u}$ needs, with $u=3x$:

$$\frac{\sin 3x}{x(x+2)}=\frac{\sin 3x}{3x}\cdot\frac{3}{x+2}\longrightarrow 1\cdot\frac{3}{2}=\boxed{\frac{3}{2}}.$$

**Check.** Substitute a value close to the target: in (b), $x=2.01$ gives 0.08308; in (c), $x=0.01$ gives 1.4923; in (a), $f(1.01)=-0.970$, $f(0.01)=-298$, $f(-0.01)=302$ and $f(1000)=1.997$.

**Pitfalls**

- Reading $\frac{0}{0}$ as "no limit". It only says that this form needs simplifying first.
- Treating $\frac{3}{0}$ as 0 or as $\infty$ without the signs. In (a) the two sides go to $-\infty$ and $+\infty$, so there is no limit at 0.
- Using $\frac{\sin 3x}{x} \to 1$. The standard limit needs the same $u$ on the top and the bottom; here it gives 3.

**In the notes.** Unit 7: Limits at a point, and Unit 7: One-sided limits, infinite limits, and limits at infinity.

**Practise it.** the Review set on this method

## 7.2 Continuity and differentiability at a join

**Problem.** Decide whether a function, often defined piecewise, is continuous or differentiable at a point; choose constants that make it so.

**Method**

1. Find $f(a)$, from the piece that includes $x = a$.
2. Find $\lim_{x\to a} f(x)$. At a join, find the two one-sided limits, each from the piece on its own side.
3. $f$ is continuous at $a$ when the limit exists and equals $f(a)$. Equate them; with an unknown constant, this is an equation for it.
4. For differentiability, impose continuity first: a function differentiable at $a$ is continuous there.
5. Find the one-sided limits of the difference quotient $\dfrac{f(a+\delta x)-f(a)}{\delta x}$ as $\delta x \to 0^-$ and as $\delta x \to 0^+$, each side from its own piece.
6. $f$ is differentiable at $a$ exactly when they are finite and agree, and $f'(a)$ is the common value. If they differ, the graph has a corner; if either is infinite, there is no derivative.

**Example.** (a) Find $a$ and $b$ so that $f(x) = \begin{cases} x^2+1, & x \le 1 \\ \dfrac{a}{x} + b, & x > 1 \end{cases}$ is differentiable at $x = 1$. (b) Show that $h(x) = \begin{cases} x^2, & x \le 0 \\ \sin x, & x > 0 \end{cases}$ is continuous but not differentiable at 0.

(a) $f(1) = 2$, from the left piece, and the one-sided limits are 2 from below and $a + b$ from above. Continuity needs $a + b = 2$. With $b = 2 - a$ the difference quotients are

$$\frac{(1+\delta x)^2+1-2}{\delta x} = 2+\delta x \xrightarrow[\delta x\to 0^-]{} 2,$$
$$\frac{1}{\delta x}\left(\frac{a}{1+\delta x} - a\right) = \frac{-a}{1+\delta x} \xrightarrow[\delta x\to 0^+]{} -a.$$

They agree only if $-a = 2$, and then $b = 2 - a$: $\boxed{a = -2,\ b = 4}$.

(b) $h(0) = 0$, and the one-sided limits are $0^2 = 0$ and $\sin 0 = 0$: $h$ is continuous at 0. The difference quotients are $\dfrac{(\delta x)^2}{\delta x} = \delta x \to 0$ from below and $\dfrac{\sin \delta x}{\delta x} \to 1$ from above. They differ,

so $h$ is *not differentiable* at 0: its graph has a corner there.

**Check.** In (a) the right piece is $4 - \frac{2}{x}$. As $x \to 1^+$ its value tends to 2 and its slope $\frac{2}{x^2}$ to 2, the same as the value and the slope $2x = 2$ of $x^2 + 1$ at 1. In (b) the slopes of the two pieces at 0 are $2x = 0$ and $\cos x = 1$: they differ, so there is a corner.

**Pitfalls**

- Matching the slopes and skipping continuity. With $a = -2$ and $b = 0$ the slopes match, but $f$ jumps from 2 to $-2$, so it is not even continuous.
- Taking $f(a)$ from the wrong piece. Read $\leq$ and $<$ in the definition.
- Concluding from continuity that $f$ is differentiable. The function $h$ in (b) is continuous and has a corner.

**In the notes.** Unit 7: Continuity, and Unit 7: Notation, higher derivatives, and a function that fails.

**Practise it.** the Review set on this method

## 7.3 Differentiating from first principles

**Problem.** Find $f'(x)$ from the definition $f'(x) = \lim_{\delta x \to 0} \dfrac{f(x+\delta x) - f(x)}{\delta x}$, without the rules.

**Method**

1. Write the difference quotient $\dfrac{f(x+\delta x) - f(x)}{\delta x}$.
2. Simplify the numerator: expand a power; bring fractions to a common denominator; for a root, multiply top and bottom by the conjugate.
3. Every term of the numerator now carries a factor $\delta x$. Cancel it with the $\delta x$ below.
4. Let $\delta x \to 0$ in what is left: "letting $\delta x \to 0$ leaves $f'(x) = \dots$".

**Example.** Differentiate from first principles (a) $f(x) = \dfrac{1}{x^2}$ and (b) $g(x) = \sqrt{x^2+4}$.

(a) Bring the two fractions to the denominator $x^2(x+\delta x)^2$ and expand the top:

$$\begin{aligned} \frac{f(x+\delta x) - f(x)}{\delta x} &= \frac{x^2 - (x+\delta x)^2}{\delta x\, x^2 (x+\delta x)^2} \\ &= \frac{-2x\,\delta x - (\delta x)^2}{\delta x\, x^2 (x+\delta x)^2} \\ &= \frac{-(2x+\delta x)}{x^2(x+\delta x)^2}. \end{aligned}$$

Letting $\delta x \to 0$ leaves $f'(x) = \dfrac{-2x}{x^4}$, so $\boxed{f'(x) = -\dfrac{2}{x^3}}$.

(b) Multiply top and bottom by $\sqrt{(x+\delta x)^2+4} + \sqrt{x^2+4}$. The top becomes $(x+\delta x)^2 + 4 - (x^2+4) = 2x\,\delta x + (\delta x)^2$, so

$$\frac{g(x+\delta x) - g(x)}{\delta x} = \frac{2x + \delta x}{\sqrt{(x+\delta x)^2+4} + \sqrt{x^2+4}}.$$

Letting $\delta x \to 0$ leaves $\dfrac{2x}{2\sqrt{x^2+4}}$, so $\boxed{g'(x) = \dfrac{x}{\sqrt{x^2+4}}}$.

**Check.** The rules give the same: $\frac{\mathrm{d}}{\mathrm{d}x}\,x^{-2} = -2x^{-3}$, and the chain rule gives $\frac{1}{2}(x^2+4)^{-1/2}\cdot 2x$.

**Pitfalls**

- Expanding $(x+\delta x)^2$ as $x^2 + (\delta x)^2$. The cross term $2x\,\delta x$ is the one that survives.
- Letting $\delta x \to 0$ before cancelling. That gives $\frac{0}{0}$ every time.
- Writing $f(x) + \delta x$ for $f(x+\delta x)$. In (b), $g(x+\delta x) = \sqrt{(x+\delta x)^2+4}$, not $\sqrt{x^2+4} + \delta x$.

**In the notes.** Unit 7: The derivative.

**Practise it.** the Review set on this method

## 7.4 Choosing between the product, quotient and chain rules

**Problem.** Differentiate a function built from standard functions by products, quotients and composition, and simplify the result.

**Method**

1. Find the last operation in the formula: a product $uv$, a quotient $u/v$, or a function of a function $f(g(x))$. It decides the rule.
2. Name $u$ and $v$; or the outer function $f(u)$ and the inner $u = g(x)$.
3. Find $u'$ and $v'$ from the standard derivatives, using the chain rule inside when $u$ or $v$ is itself a function of a function.
4. Apply the rule: $(uv)' = u'v + uv'$; $\left(\frac{u}{v}\right)' = \dfrac{u'v - uv'}{v^2}$; $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u}\cdot\frac{\mathrm{d}u}{\mathrm{d}x}$.
5. Simplify: clear any fraction inside the numerator, and take out the common factor or power.

**Variants.** A *tangent line* at $x = a$: find $f(a)$ and $f'(a)$, write the point–slope form $y - f(a) = f'(a)(x-a)$, and rearrange to $y = mx + c$. *Motion* along a line: the velocity is $v = \dot{s}$ and the acceleration $a = \ddot{s}$, with units; the body is at rest where $v = 0$, found by factoring, and the sign of $v$ gives the direction. *Higher derivatives*: differentiate again and again, listing $f'$, $f''$, …; for a polynomial, stop at the first 0. A *sensitivity*: evaluate $\frac{\mathrm{d}L}{\mathrm{d}w}$ at the current $w$; a small step that reduces $L$ moves $w$ against the sign of $\frac{\mathrm{d}L}{\mathrm{d}w}$.

**Example.** (a) Differentiate $y = x\sqrt{5-x}$, and find the tangent line at $x = 4$. (b) Differentiate $y = \dfrac{\cos x}{1+\sin x}$.

(a) A product: $u = x$ and $v = (5-x)^{1/2}$. Then $u' = 1$, and by the chain rule, with inner $5-x$ of derivative $-1$, $v' = \frac{1}{2}(5-x)^{-1/2}\cdot(-1)$. So

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \sqrt{5-x} - \frac{x}{2\sqrt{5-x}}.$$

Put both terms over the common denominator $2\sqrt{5-x}$; the top is $2(5-x)-x$:

$$\boxed{\frac{dy}{dx} = \frac{10-3x}{2\sqrt{5-x}}}$$

At $x=4$: $y = 4\cdot 1 = 4$ and $\frac{dy}{dx} = \frac{-2}{2} = -1$. The point–slope form $y-4 = -(x-4)$ gives the tangent $\boxed{y = 8-x}$.

(b) A quotient: $u = \cos x$ and $v = 1+\sin x$, with $u' = -\sin x$ and $v' = \cos x$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{-\sin x\,(1+\sin x) - \cos^2 x}{(1+\sin x)^2} \\ &= \frac{-\sin x - (\sin^2 x + \cos^2 x)}{(1+\sin x)^2} \\ &= \frac{-(1+\sin x)}{(1+\sin x)^2},\end{aligned}$$

so $\boxed{\dfrac{dy}{dx} = -\dfrac{1}{1+\sin x}}$.

**Check.** The slope of a short secant centred on the point tests the derivative itself. In (a), $\dfrac{y(4.001)-y(3.999)}{0.002} \approx -1.000$, the slope $-1$ at $x=4$. In (b), $\dfrac{y(0.001)-y(-0.001)}{0.002} \approx -1.000$, and $-\frac{1}{1+0} = -1$.

**Pitfalls**

- Writing $(uv)' = u'v'$. In (a) it gives $-\frac{1}{2}$ at $x=4$ instead of $-1$.
- Dropping the inner derivative $-1$ of $5-x$. It gives $\frac{10-x}{2\sqrt{5-x}}$, which is 3 at $x=4$, and the wrong tangent.
- Turning the quotient rule round: $uv' - u'v$ over $v^2$ gives $+\frac{1}{1+\sin x}$ in (b), the right size with the wrong sign.

**In the notes.** Unit 7: The sum, product and quotient rules, Unit 7: The chain rule, and Unit 7: Three readings of the derivative, for the tangent line, the motion and the sensitivity.

**Practise it.** the Review set on this method

## 7.5 Implicit differentiation

**Problem.** A curve is given by an equation in $x$ and $y$. Find the slope $y' = \frac{dy}{dx}$, its value at a point, and the second derivative $y''$.

**Method**

1. Differentiate both sides with respect to $x$, treating $y$ as a function of $x$. By the chain rule $\frac{d}{dx}\,g(y) = g'(y)\,y'$; a product such as $xy$ needs the product rule, $y + xy'$.
2. Collect the terms in $y'$ on one side.
3. Solve for $y'$, where its coefficient is not zero.
4. Evaluate at the point, having checked that the point lies on the curve.
5. For $y''$, differentiate the relation from step 1 once more, substitute the values of $x$, $y$ and

$y'$, and solve for $y''$.

**Example.** For the curve $e^y + xy = e$, find $y'$, and find $y'$ and $y''$ at the point $(0, 1)$.

The point lies on the curve: $e^1 + 0 = e$. Differentiate, with the chain rule on $e^y$ and the product rule on $xy$:

$$e^y y' + y + xy' = 0 \quad \Longrightarrow \quad \boxed{y' = -\frac{y}{e^y + x}}$$

wherever $e^y + x \neq 0$. At $(0, 1)$ this gives $\boxed{y' = -\frac{1}{e}}$.

Differentiate the relation $e^y y' + y + xy' = 0$ again (product rule on $e^y y'$ and on $xy'$):

$$e^y (y')^2 + e^y y'' + y' + y' + xy'' = 0.$$

At $(0, 1)$ with $y' = -\frac{1}{e}$ this reads $\frac{1}{e} - \frac{2}{e} + e\,y'' = 0$, so $\boxed{y'' = \frac{1}{e^2}}$.

**Check.** This equation can be solved for $x$ instead: $x = \dfrac{e - e^y}{y}$, with $\frac{dx}{dy} = \dfrac{-ye^y - (e - e^y)}{y^2}$, which is $-e$ at $y = 1$. The slope is $\frac{dy}{dx} = 1 \big/ \frac{dx}{dy} = -\frac{1}{e}$.

**Pitfalls**

- Writing $\frac{d}{dx} e^y = e^y$. The chain rule adds the factor $y'$.
- Differentiating $xy$ as $xy'$ alone. The product rule gives $y + xy'$.
- Substituting the point before differentiating. With $x = 0$ put in first, the $xy$ term has vanished, and $e^y y' = 0$ gives the false slope 0.

**In the notes.** Unit 7: Implicit differentiation.

**Practise it.** the Review set on this method

## 7.6 Parametric differentiation

**Problem.** A curve is given by $x = x(t)$ and $y = y(t)$. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$, and the points with a horizontal or a vertical tangent.

**Method**

1. Find $dx/dt$ and $dy/dt$. (Dots are for derivatives with respect to time only.)
2. State where $dx/dt \neq 0$: the slope formula holds only there.
3. Divide, $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$, and simplify.
4. For $\frac{d^2y}{dx^2}$, differentiate $\frac{dy}{dx}$ with respect to $t$ and divide by $dx/dt$ again.
5. The tangent is horizontal where $dy/dt = 0$ and $dx/dt \neq 0$, and vertical where $dx/dt = 0$ and $dy/dt \neq 0$. Give the points $(x, y)$, not just the values of $t$.

**Example.** The figure-of-eight curve $x = 2\cos t$, $y = \sin 2t$ is traced once for $0 \leq t < 2\pi$. (a) Find $\frac{dy}{dx}$. (b) Find the points with a horizontal or a vertical tangent. (c) Find $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$.

(a) $\dfrac{dx}{dt} = -2\sin t$ and $\dfrac{dy}{dt} = 2\cos 2t$, so for $\sin t \neq 0$,

$$\boxed{\frac{dy}{dx} = \frac{2\cos 2t}{-2\sin t} = -\frac{\cos 2t}{\sin t}}$$

(b) *Horizontal*: $\cos 2t = 0$ at $t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$, where $\sin t \neq 0$. They give $\boxed{(\pm\sqrt{2},\ \pm 1)}$, all four sign choices. *Vertical*: $\sin t = 0$ at $t = 0$ and $t = \pi$, where $\dfrac{dy}{dt} = 2 \neq 0$. They give $\boxed{(\pm 2,\ 0)}$.

(c) By the quotient rule,

$$\frac{d}{dt}\left(-\frac{\cos 2t}{\sin t}\right) = \frac{2\sin 2t\,\sin t + \cos 2t\,\cos t}{\sin^2 t}.$$

At $t = \frac{\pi}{4}$, where $\sin 2t = 1$, $\cos 2t = 0$ and $\sin t = \frac{\sqrt{2}}{2}$, this is $\dfrac{\sqrt{2}}{1/2} = 2\sqrt{2}$. Divide by $\dfrac{dx}{dt} = -\sqrt{2}$: $\boxed{\dfrac{d^2y}{dx^2} = -2}$. The curve is concave down at $(\sqrt{2}, 1)$.

**Check.** At $t = \frac{\pi}{4}$, $y = \sin 2t$ takes its largest value 1, so a horizontal tangent there, with the curve bending down, is expected. Eliminating $t$ on $0 < t < \pi$, where $\sin t > 0$, gives $y = \frac{1}{2}x\sqrt{4 - x^2}$; its derivative at $x = \sqrt{2}$ is 0 and its second derivative there is $-2$.

**Pitfalls**

- Dividing the second derivatives, $\dfrac{d^2y/dt^2}{d^2x/dt^2}$. At $t = \frac{\pi}{4}$ that gives $\frac{-4}{-\sqrt{2}} = 2\sqrt{2}$, the wrong size and the wrong sign.
- Dividing the wrong way round, $(dx/dt)/(dy/dt)$. That is $\frac{dx}{dy}$.
- Calling a point horizontal without checking $dx/dt \neq 0$ there. Where both vanish, the formula says nothing.

**In the notes.** Unit 7: Parametric differentiation.
**Practise it.** the Review set on this method

## 7.7 Logarithmic differentiation

**Problem.** Differentiate a function with the variable in both a base and an exponent, or a product or quotient of many factors.

**Method**

1. Take $\ln|\cdot|$ of both sides. Where $y > 0$ throughout, as for $x^x$ with $x > 0$, plain ln will do; say so.
2. Simplify with the log laws: a product becomes a sum, a quotient a difference, and a power a multiplier.
3. Differentiate both sides with respect to $x$. The left side gives $\dfrac{1}{y}\dfrac{dy}{dx}$ by the chain rule; the right side goes term by term.
4. Multiply back by $y$, written in terms of $x$, and state where the result holds: away from the zeros of the factors. At a zero of a power factor, such as $x = 0$ for $x^3$, $y$ is still defined:

find $y'$ there by the product rule instead.

**Example.** (a) Differentiate $y = x^{\sqrt{x}}$ for $x > 0$, and evaluate $\frac{dy}{dx}$ at $x = 4$. (b) Differentiate $y = \dfrac{x^3\sqrt{x+1}}{(x-2)^2}$, and evaluate $\frac{dy}{dx}$ at $x = 3$.

(a) For $x > 0$, $y > 0$, so $\ln y = \sqrt{x}\,\ln x$. Differentiate, by the product rule on the right:

$$\frac{1}{y}\frac{dy}{dx} = \frac{\ln x}{2\sqrt{x}} + \sqrt{x}\cdot\frac{1}{x} = \frac{\ln x + 2}{2\sqrt{x}}.$$

Multiply back by $y$:

$$\boxed{\frac{dy}{dx} = x^{\sqrt{x}}\,\frac{\ln x + 2}{2\sqrt{x}}}$$

At $x = 4$: $y = 4^2 = 16$, so $\frac{dy}{dx} = 16\cdot\dfrac{\ln 4 + 2}{4} = \boxed{8(1+\ln 2)}$.

(b) For $x > -1$, $x \neq 0$, $x \neq 2$: $\ln|y| = 3\ln|x| + \frac{1}{2}\ln(x+1) - 2\ln|x-2|$. Differentiate:

$$\frac{1}{y}\frac{dy}{dx} = \frac{3}{x} + \frac{1}{2(x+1)} - \frac{2}{x-2}.$$

Multiply back by $y$:

$$\frac{dy}{dx} = \frac{x^3\sqrt{x+1}}{(x-2)^2} \times \left(\frac{3}{x} + \frac{1}{2(x+1)} - \frac{2}{x-2}\right).$$

At $x = 3$, $y = \frac{27\cdot 2}{1} = 54$ and the bracket is $1 + \frac{1}{8} - 2 = -\frac{7}{8}$, so

$$\left.\frac{dy}{dx}\right|_{x=3} = 54\cdot\left(-\frac{7}{8}\right) = \boxed{-\frac{189}{4}}.$$

At $x = 0$, where $y = 0$, the product rule on $x^3\cdot\dfrac{\sqrt{x+1}}{(x-2)^2}$ gives two terms, each with a factor $x^2$, so $y'(0) = 0$.

**Check.** The slope of a short secant centred on the point. In (a), $\dfrac{y(4.001) - y(3.999)}{0.002} \approx 13.545$, and $8(1+\ln 2) \approx 13.545$. In (b), $\dfrac{y(3.001) - y(2.999)}{0.002} \approx -47.25$, and $-\frac{189}{4} = -47.25$.

**Pitfalls**

- Using the power rule on $x^{\sqrt{x}}$ as if $\sqrt{x}$ were a constant: $\sqrt{x}\,x^{\sqrt{x}-1}$ is 8 at $x = 4$, not 13.5.
- Stopping at $\dfrac{y'}{y} = \ldots$. The derivative needs the factor $y$ multiplied back.
- Taking plain ln in (b). For $-1 < x < 0$ the factor $x^3$ is negative, and $\ln x^3$ is not defined there; $\ln|x^3| = 3\ln|x|$ is.

**In the notes.** Unit 7: Logarithmic differentiation.
**Practise it.** the Review set on this method

# Unit 8 Differentiation II: applications

Seven methods. Nearly every problem starts by computing $f'$, so what decides the method is what the question asks for. The derivative of an inverse function is a variant of Method 8.2.

### Which method?

| The question asks for | Method |
|---|---|
| the tangent or the normal at a point | 8.1 |
| "show that the equation has a root in $[a, b]$" | 8.2, a sign change (IVT) |
| "… and it has exactly one root" | 8.2, add the sign of $f'$ |
| a value $c$ with $f'(c)$ equal to a secant slope; a bound on $\lvert f(b) - f(a)\rvert$ | 8.2, the mean value theorem |
| the slope of $f^{-1}$ at a point | 8.2, variant: $1/f'(a)$ |
| a limit of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ | 8.3, l'Hôpital's rule |
| a limit of the form $0 \cdot \infty$ or $\infty - \infty$ | 8.3, rewrite as a quotient first |
| a local maximum or minimum; "classify" | 8.4, the derivative tests |
| the largest or smallest value on $[a, b]$ | 8.4, the closed-interval method |
| a sketch of $y = f(x)$ | 8.5, the checklist |
| the shape of $f$ from a graph of $f'$ | 8.5, variant: read the signs |
| the largest or smallest value, from a description or on an open interval | 8.6, reduce to one variable |
| a root to six decimal places | 8.7, Newton's method |
| a minimum, by iterating with a step size $\eta$ | 8.7, gradient descent |

## 8.1 Tangent and normal

**Problem.** Find the tangent line and the normal line to the curve $y = f(x)$ at the point where $x = a$, or where the tangent has a given slope.

### Method

1. Compute $f(a)$ and $f'(a)$: differentiate first, then put $x = a$. If the point is given by its slope instead (a horizontal tangent has slope 0), first solve $f'(a) =$ that slope for $a$, keeping solutions in the domain.
2. Tangent: $y = f(a) + f'(a)\,(x - a)$. Simplify it to the form $y = mx + c$.
3. Normal: its slope is $-1/f'(a)$, so $y = f(a) - \dfrac{1}{f'(a)}\,(x - a)$. If $f'(a) = 0$, the tangent is the horizontal line $y = f(a)$ and the normal is the vertical line $x = a$.
4. Check that the two slopes multiply to $-1$ (when $f'(a) \neq 0$), and that both lines pass through $\big(a, f(a)\big)$.

**Example.** For the curve $y = x\sqrt{x+3}$, find (a) the tangent and the normal at the point where $x = 1$, and (b) the tangent and the normal at the point where the tangent is horizontal.

By the product rule,

$$f'(x) = \sqrt{x+3} + \frac{x}{2\sqrt{x+3}} = \frac{3x+6}{2\sqrt{x+3}}.$$

(a) $f(1) = 1 \cdot \sqrt{4} = 2$ and $f'(1) = 2 + \frac{1}{4} = \frac{9}{4}$. The tangent is

$$y = 2 + \tfrac{9}{4}(x-1) \quad \Longrightarrow \quad \boxed{y = \tfrac{9}{4}x - \tfrac{1}{4}}$$

The normal has slope $-1/\frac{9}{4} = -\frac{4}{9}$:

$$y = 2 - \tfrac{4}{9}(x-1) \quad \Longrightarrow \quad \boxed{y = -\tfrac{4}{9}x + \tfrac{22}{9}}$$

(b) $f'(x) = 0$ where $3x + 6 = 0$, that is at $x = -2$, which is in the domain $x \geq -3$. There $f(-2) = -2\sqrt{1} = -2$. The tangent is horizontal, so the normal is vertical:

$$\boxed{\text{tangent } y = -2, \quad \text{normal } x = -2}$$

**Check.** (a) The slopes multiply to $\frac{9}{4} \cdot \left(-\frac{4}{9}\right) = -1$, and both lines pass through $(1, 2)$: $\frac{9}{4} - \frac{1}{4} = 2$ and $-\frac{4}{9} + \frac{22}{9} = 2$. The slope itself: $\big(f(1.001) - f(1)\big)/0.001 \approx 2.2502$, close to $\frac{9}{4}$. (b) $y = -2$ and $x = -2$ both pass through $(-2, -2) = \big(-2, f(-2)\big)$.

**Pitfalls**

- Leaving $x$ in the slope. The tangent's slope is the number $f'(1) = \frac{9}{4}$, not the function $f'(x)$.
- Taking the normal's slope as $\frac{4}{9}$ or as $-\frac{9}{4}$. It is minus the reciprocal, $-1/f'(a)$.
- Writing the normal at a horizontal tangent as $y = \ldots$. Its slope $-1/0$ does not exist; the normal is the line $x = a$.

**In the notes.** Unit 8: Tangent and normal lines.
**Practise it.** the Review set on this method

## 8.2 A root exists and is unique (IVT and monotonicity); the mean value theorem

**Problem.** Show that an equation has a root in a given interval, and that it has no other; find the point $c$ that the mean value theorem promises, or use the theorem to bound a difference.

**Method**

1. For a root: write the equation as $f(x) = 0$, with everything on one side.
2. State that $f$ is continuous on $[a, b]$: polynomials, $e^x$, $\ln x$, $\sin x$, $\cos x$ and their sums, products, quotients and compositions are continuous wherever they are defined, so check that $f$ is defined at every point of $[a, b]$.
3. Evaluate $f(a)$ and $f(b)$, and show that they have opposite signs.
4. Conclude by the intermediate value theorem: $f(c) = 0$ for some $c \in (a, b)$.
5. Uniqueness: show that $f' > 0$ (or $f' < 0$) on the whole domain of $f$, which must be an interval, usually $\mathbb{R}$. By the monotonicity corollary, which holds on an interval, $f$ is then strictly monotone, so it takes the value 0 at most once.
6. Mean value theorem (start here for a secant slope or a bound): check that $f$ is continuous

on $[a, b]$ and differentiable on $(a, b)$. Compute the secant slope $\dfrac{f(b) - f(a)}{b - a}$, solve $f'(c) =$ slope, and keep only the solutions with $c \in (a, b)$.

**Variants.** A *bound*: apply the mean value theorem between the two points, $f(b) - f(a) = f'(c)\,(b - a)$, and replace $|f'(c)|$ by the largest value $|f'|$ takes between the two points. The *derivative of an inverse*: once $f' > 0$ (or $f' < 0$) shows that $f$ is one-to-one, $(f^{-1})'(b) = 1/f'(a)$, where $a$ solves $f(a) = b$ (usually by inspection) and $f'(a) \neq 0$. No formula for $f^{-1}$ is needed.

**Example.** (a) Show that $e^x + x = 2$ has exactly one real solution, and that it lies in $(0, 1)$. (b) Find the value of $c$ that the mean value theorem gives for $f(x) = \dfrac{1}{x}$ on $[1, 4]$.

(a) Write the equation as $g(x) = e^x + x - 2 = 0$. The function $g$, an exponential plus a polynomial, is continuous on $\mathbb{R}$, and

$$g(0) = 1 + 0 - 2 = -1 < 0, \qquad g(1) = e - 1 > 0.$$

By the intermediate value theorem, $g(c) = 0$ for some $c \in (0, 1)$. And $g'(x) = e^x + 1 > 0$ for every $x$, so $g$ is strictly increasing and has at most one zero. So there is $\boxed{\text{exactly one root, in } (0, 1)}$.

(b) $f(x) = 1/x$ is continuous on $[1, 4]$ and differentiable on $(1, 4)$, since 0 is not in the interval. The secant slope is

$$\frac{f(4) - f(1)}{4 - 1} = \frac{\frac{1}{4} - 1}{3} = -\frac{1}{4}.$$

Solve $f'(c) = -\dfrac{1}{c^2} = -\dfrac{1}{4}$: $c^2 = 4$, so $c = \pm 2$. Only one of them lies in $(1, 4)$: $\boxed{c = 2}$.

**Check.** (a) One more sign: $g(\frac{1}{2}) = \sqrt{e} - \frac{3}{2} \approx 0.15 > 0$, so the root is in fact in $(0, \frac{1}{2})$, and Method 8.7 finds it there. (b) $f'(2) = -\frac{1}{4}$, the slope of the secant.

**Pitfalls**

- Stopping at the sign change when "exactly one" is asked. The intermediate value theorem gives at least one root, never uniqueness.
- Skipping the continuity check. On $[-1, 1]$ the function $1/x$ has $f(-1) < 0 < f(1)$ but no root. Its secant slope is 1 while $f' < 0$ everywhere, so there is no mean-value point either: $1/x$ is not defined at 0, so it is not continuous on $[-1, 1]$.
- Keeping $c = -2$. The theorem promises a point strictly inside $(a, b)$.

**In the notes.** Unit 8: Two theorems for continuous functions, Unit 8: The Mean Value Theorem, and Unit 8: The derivative of an inverse function.

**Practise it.** the Review set on this method For the derivative of an inverse, the Review set on this variant

## 8.3 l'Hôpital's rule

**Problem.** Evaluate a limit of an indeterminate form: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$ or $\infty - \infty$.

**Method**

1. Substitute the limit point and name the form. If it is not indeterminate, the rule does not apply: a finite value is the answer, and a non-zero number over 0 means an infinite limit,

or none if the two sides differ (Unit 7).

2. Rewrite $0 \cdot \infty$ as a quotient, $fg = \dfrac{f}{1/g}$, sending down a factor $g$ whose reciprocal is simple (for $g = x$, $1/x$); if the new quotient is no simpler, send down the other factor. Rewrite a difference of fractions over a common denominator; with no fractions, factor out the larger term: $x - \ln x = x\left(1 - \frac{\ln x}{x}\right)$. Each leaves a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ quotient.
3. Apply the rule: differentiate the numerator and the denominator separately, and mark the step $\overset{\text{H}}{=}$.
4. Simplify, and check the form again before every repeat. Stop using the rule as soon as the form is no longer indeterminate.
5. Evaluate, as in step 1.

**Example.** Evaluate (a) $\lim_{x\to\infty} x\left(e^{2/x} - 1\right)$ and (b) $\lim_{x\to 0}\left(\dfrac{1}{x} - \dfrac{1}{e^x - 1}\right)$.

(a) As $x \to \infty$, $e^{2/x} \to 1$, so the form is $\infty \cdot 0$. Rewrite it as $\dfrac{e^{2/x} - 1}{1/x}$, a $\frac{0}{0}$ form. By the chain rule the top differentiates to $-\frac{2}{x^2}e^{2/x}$ and the bottom to $-\frac{1}{x^2}$, so

$$\lim_{x\to\infty} \frac{e^{2/x} - 1}{1/x} \overset{\text{H}}{=} \lim_{x\to\infty} 2e^{2/x} = \boxed{2}.$$

(b) The form is $\infty - \infty$. Over the common denominator $x(e^x - 1)$,

$$\frac{1}{x} - \frac{1}{e^x - 1} = \frac{e^x - 1 - x}{x(e^x - 1)},$$

a $\frac{0}{0}$ form. Then

$$\lim_{x\to 0} \frac{e^x - 1 - x}{x(e^x - 1)} \overset{\text{H}}{=} \lim_{x\to 0} \frac{e^x - 1}{e^x - 1 + xe^x},$$

which is still $\frac{0}{0}$, so apply the rule again:

$$\lim_{x\to 0} \frac{e^x - 1}{e^x - 1 + xe^x} \overset{\text{H}}{=} \lim_{x\to 0} \frac{e^x}{2e^x + xe^x} = \boxed{\tfrac{1}{2}}.$$

**Check.** Numerically: at $x = 1000$ the expression in (a) is 2.0020, and at $x = 0.01$ the one in (b) is 0.4992.

**Pitfalls**

- Using the rule once too often. In (b) the last quotient tends to $\frac{1}{2}$, not $\frac{0}{0}$; one more application would give $\frac{1}{3}$.
- Applying the rule to $0 \cdot \infty$ or $\infty - \infty$ before rewriting it as a single quotient.
- Differentiating $f/g$ by the quotient rule. The rule differentiates the top and the bottom separately.

**In the notes.** Unit 8: l'Hôpital's rule.

**Practise it.** the Review set on this method

## 8.4 Classifying critical points; the closed-interval method

**Problem.** Find the critical points of $f$ and classify each as a local maximum, a local minimum or neither; find the absolute maximum and minimum of a continuous $f$ on a closed interval $[a, b]$.

**Method**

1. Find $f'$ and factor it. The critical points are the solutions of $f'(x) = 0$ in the domain, together with any point of the domain where $f'$ does not exist.
2. Classify each stationary point by the second-derivative test where it is quick: $f''(c) > 0$ gives a local minimum, $f''(c) < 0$ a local maximum.
3. If $f''(c) = 0$, or $f'(c)$ does not exist, use the first-derivative test instead: the sign of $f'$ on each side of $c$. From + to − is a maximum, from − to + a minimum, and the same strict sign on both sides is neither. If $f' = 0$ on one side, as for $(\mathrm{ReLU}(x))^2$ at 0, no pattern applies: compare $f(x)$ with $f(c)$ on both sides.
4. State each result as a point: "local maximum at $(c, f(c))$", or neither.
5. On a closed interval: keep the critical points inside $[a, b]$, and evaluate $f$ there and at both endpoints.
6. The largest of these values is the absolute maximum, and the smallest is the absolute minimum.

**Example.** Let $f(x) = x^4 - 4x^3$. (a) Find and classify its critical points. (b) Find its absolute maximum and minimum on $[-1, 4]$.

(a) $f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$, which is zero at $x = 0$ and $x = 3$ and exists everywhere. With $f''(x) = 12x^2 - 24x$,

$$f''(3) = 108 - 72 = 36 > 0 \;\Rightarrow\; \text{local min at } (3, -27).$$

At $x = 0$, $f''(0) = 0$ and the test is inconclusive. The factor $4x^2$ is positive on both sides of 0 and $x - 3$ is negative near 0, so $f' < 0$ on both sides: $f$ keeps decreasing through $x = 0$. So

$$\boxed{\text{local min at } (3, -27)}, \qquad \boxed{\text{neither at } (0, 0)},$$

the second a horizontal point of inflection.

(b) Both critical points lie in $[-1, 4]$. The values there and at the endpoints are

$$f(-1) = 5, \quad f(0) = 0, \quad f(3) = -27, \quad f(4) = 0.$$

So $\boxed{\text{abs. max 5 at } x = -1}$ and $\boxed{\text{abs. min } -27 \text{ at } x = 3}$.

**Check.** $f(x) + 27 = (x-3)^2(x^2 + 2x + 3)$, and $x^2 + 2x + 3 = (x+1)^2 + 2 > 0$, so $f(x) \geq -27$ for every $x$, with equality only at $x = 3$. And the signs used at 0: $f'(-1) = -16$ and $f'(1) = -8$.

**Pitfalls**

- Reading $f''(0) = 0$ as "an inflection" or "neither" without looking further: $x^4$ also has $f''(0) = 0$, at a minimum. Fall back on the sign of $f'$.
- Skipping the endpoints. Here the absolute maximum, 5 at $x = -1$, is at an endpoint, not at a critical point.
- Dividing $4x^2(x - 3) = 0$ by $x^2$, which loses the critical point $x = 0$.

**In the notes.** Unit 8: Critical points and the first-derivative test, Unit 8: Concavity and the second-derivative test, and Unit 8: Two theorems for continuous functions, for the extreme value theorem.

**Practise it.** the Review set on this method

### 8.5 Sketching a curve by the checklist, or from a graph of $f'$

**Problem.** Sketch $y = f(x)$ for a polynomial or a rational function, showing its intercepts, asymptotes, turning points and inflections; or describe $f$ from a graph of $f'$ alone.

**Method**

1. *Domain and symmetry*: where $f$ is defined; whether it is even or odd.
2. *Intercepts*: $f(0)$, if 0 is in the domain, and the solutions of $f(x) = 0$.
3. *Asymptotes*, for a rational function: cancel any common factor of top and bottom first (a zero that cancels completely leaves a hole); then vertical where the denominator vanishes and the numerator does not; horizontal from the limits as $x \to \pm\infty$.
4. *Increase and decrease*: the sign of $f'$ on each interval between its zeros and the gaps in the domain. Classify the stationary points (Method 8.4).
5. *Concavity and inflections*: the sign of $f''$. An inflection is a sign change of $f''$ at a point of the domain.
6. Describe the shape in a sentence, then draw it, labelling the points you found.

**Variants.** From a graph of $f'$: $f$ increases where $f' > 0$ and decreases where $f' < 0$. A zero where $f'$ changes from + to − is a local maximum of $f$; one where it changes from − to +, a local minimum. The slope of the graph of $f'$ is $f''$: where that graph rises, $f$ is concave up, where it falls, concave down, and each turning point of $f'$ is an inflection of $f$.

**Example.** Find the features needed to sketch $f(x) = \dfrac{x-1}{x^2}$, and describe the curve.

- *Domain and symmetry.* $\mathbb{R} \setminus \{0\}$. Neither even nor odd: $f(-1) = -2$ but $f(1) = 0$.
- *Intercepts.* None on the $y$-axis, since 0 is not in the domain. $f(x) = 0$ only at $x = 1$.
- *Asymptotes.* At $x = 0$ the denominator vanishes and the numerator is $-1$, so $x = 0$ is a vertical asymptote, with $f \to -\infty$ on both sides since $x^2 > 0$. And $f(x) = \frac{1}{x} - \frac{1}{x^2} \to 0$ as $x \to \pm\infty$, so the asymptotes are $\boxed{x = 0 \text{ and } y = 0}$.
- *Increase/decrease.* By the quotient rule, $f'(x) = \dfrac{2-x}{x^3}$. Its sign is −, +, − on $(-\infty, 0)$, $(0, 2)$ and $(2, \infty)$: $f$ decreases, increases, then decreases.
- *Classification.* The only stationary point is $x = 2$, where $f'$ changes from + to −: a local maximum at $\boxed{(2, \frac{1}{4})}$.
- *Concavity.* $f''(x) = \dfrac{2x-6}{x^4}$ has the sign of $x - 3$: concave down on $(-\infty, 0)$ and on $(0, 3)$, concave up on $(3, \infty)$, with an inflection at $\boxed{(3, \frac{2}{9})}$.

The left branch starts just below $y = 0$ far to the left and falls to $-\infty$ beside $x = 0$. The right branch rises from $-\infty$ beside $x = 0$, crosses the axis at $(1, 0)$, peaks at $(2, \frac{1}{4})$, changes its bend

at $(3, \frac{2}{9})$ and falls back towards $y = 0$, staying above it.

**Check.** $\frac{1}{4} - f(x) = \dfrac{(x-2)^2}{4x^2} \geq 0$, so $(2, \frac{1}{4})$ is the highest point of the whole graph, as the description says. The sign of $f$ agrees too: negative for $x < 1$, positive for $x > 1$.

**Pitfalls**

- Reading a sign change across an asymptote as a turning point. Here $f'$ changes from $-$ to $+$ across $x = 0$, but $x = 0$ is not in the domain, so it is no minimum.
- Giving a $y$-intercept when 0 is not in the domain.
- Putting a branch on the wrong side of the horizontal asymptote $y = L$. The sign of $f(x) - L$ for large $|x|$ decides it. Here $L = 0$, so it is the sign of $f$: below $y = 0$ on the left, above on the right.

**In the notes.** Unit 8: Curve sketching, with its two full sketches, and Unit 8: Concavity and the second-derivative test.

**Practise it.** the Review set on this method

## 8.6 An optimisation word problem

**Problem.** A quantity described in words is to be made as large or as small as possible, subject to a constraint.

**Method**

1. Name the quantity $Q$ to be optimised and the variables it depends on. A labelled sketch helps.
2. Use the constraint to write $Q$ as a function of a single variable, and state the domain: the physically meaningful interval.
3. Find the critical points: solve $Q' = 0$, and add any interior point of the domain where $Q'$ does not exist.
4. Identify the optimum by the second-derivative test or the sign of $Q'$. Compare it with the endpoints of a closed interval, or with the behaviour at the ends of an open one.
5. Answer the question asked: every dimension and the optimal value, with units.

**Example.** An open-topped box with a square base must hold $500\,\text{cm}^3$. What dimensions use the least card, and how much card is that?

The quantity to minimise is the area $A$ of card. Let the base have side $x$ and the box height $h$. The base and the four sides give $A = x^2 + 4xh$. The volume $x^2h = 500$ fixes $h = 500/x^2$, so

$$A(x) = x^2 + 4x \cdot \frac{500}{x^2} = x^2 + \frac{2000}{x}, \qquad x > 0.$$

Then

$$A'(x) = 2x - \frac{2000}{x^2} = 0 \implies x^3 = 1000, \quad x = 10.$$

$A''(x) = 2 + \dfrac{4000}{x^3} > 0$ for all $x > 0$, so this is a minimum; and $A \to \infty$ both as $x \to 0^+$ and as

$x \to \infty$, so it is the global minimum. The height is $h = 500/100 = 5$:

$$\boxed{10 \times 10 \times 5 \text{ cm}}, \qquad \boxed{A_{\min} = 300 \text{ cm}^2}.$$

**Check.** $10^2 \cdot 5 = 500$. Neighbouring widths cost more: $A(9) \approx 303.2$ and $A(11) \approx 302.8$. Exactly, $A(x) - 300 = \dfrac{(x-10)^2(x+20)}{x} \geq 0$ for $x > 0$.

**Pitfalls**

- Counting a lid: $2x^2 + 4xh$ is the card for a closed box.
- Stopping at the second-derivative test on an open domain. It shows a local minimum; the behaviour at the ends shows that it is the global one.
- Stopping at $x = 10$. The question asks for every dimension and for the amount of card.

**In the notes.** Unit 8: Solving optimisation problems, with the most economical can.
**Practise it.** the Review set on this method

## 8.7 Newton's method and gradient descent

**Problem.** Find a root of $f(x) = 0$ to a given number of decimal places by Newton's method; minimise $f$ numerically by gradient descent with a learning rate $\eta$.

**Method**

1. Newton: write $f$, $f'$ and the iteration $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$ explicitly, with $f$ and $f'$ substituted.
2. Start from $x_0$ near the root: a point of the interval that the intermediate value theorem gives (an endpoint will do), with $f'(x_0) \neq 0$. Tabulate $x_n$ and $f(x_n)$ to six decimal places. If the iterates cycle or leave the interval, halve it by the sign at its midpoint and start again there.
3. Stop when two iterates agree to the places asked for, and confirm the rounding by a sign change: for six places, $f$ changes sign between $x - 0.0000005$ and $x + 0.0000005$, where $x$ is the answer. Comment on the speed: near a simple root the number of correct digits roughly doubles each step.
4. Gradient descent: find $f'$, write the update $x_{n+1} = x_n - \eta f'(x_n)$ in simplified form, and iterate from $x_0$.
5. For a quadratic with its minimum at $x^*$ and $k = f''$, the error $e_n = x_n - x^*$ obeys $e_{n+1} = (1 - \eta k)\, e_n$. The iterates converge exactly when $0 < \eta < 2/k$.

**Why it works.** $x_{n+1}$ is where the tangent at $x_n$ meets the axis: put $y = 0$ in $y = f(x_n) + f'(x_n)(x - x_n)$ (Method 8.1).

**Example.** (a) Method 8.2 showed that $\mathrm{e}^x + x - 2 = 0$ has exactly one root, in $(0, 1)$. Find it to six decimal places by Newton's method from $x_0 = 0$. (b) Minimise $f(x) = 3x^2 - 6x + 2$ by gradient descent with $\eta = 0.25$ from $x_0 = 3$: compute $x_1$, $x_2$, $x_3$, and find the values of $\eta$ for which the iteration converges.

(a) With $f(x) = \mathrm{e}^x + x - 2$ and $f'(x) = \mathrm{e}^x + 1$,

$$x_{n+1} = x_n - \frac{\mathrm{e}^{x_n} + x_n - 2}{\mathrm{e}^{x_n} + 1}.$$

The first step is $x_1 = 0 - \frac{-1}{2} = 0.5$, and then

| $n$ | $x_n$ | $f(x_n)$ |
|---|---|---|
| 0 | 0.000000 | −1.000000 |
| 1 | 0.500000 | 0.148721 |
| 2 | 0.443852 | 0.002551 |
| 3 | 0.442855 | 0.000001 |
| 4 | 0.442854 | 0.000000 |
| 5 | 0.442854 | 0.000000 |

The last two iterates, $x_4$ and $x_5$, agree to six places: $\boxed{x \approx 0.442854}$. The errors run 0.44, 0.057, 0.0010, $3 \times 10^{-7}$: the number of correct digits roughly doubles each step.

(b) $f'(x) = 6x - 6$, so $x_{n+1} = x_n - 0.25(6x_n - 6) = -0.5\,x_n + 1.5$, and

$$\boxed{x_1 = 0,\ x_2 = 1.5,\ x_3 = 0.75}$$

The minimum is at $x^* = 1$, where $f' = 0$. Here $k = f'' = 6$, so $e_{n+1} = (1 - 0.25 \cdot 6)\, e_n = -0.5\, e_n$: the error halves and changes sign each step, 2, −1, 0.5, −0.25. The iteration converges exactly when $0 < \eta < 2/k$, that is $\boxed{0 < \eta < \frac{1}{3}}$.

**Check.** (a) $f(0.4428535) < 0 < f(0.4428545)$, so the root rounds to 0.442854. (b) With $\eta = 0.4$, outside the range, the update is $x_{n+1} = -1.4\,x_n + 2.4$, and $3 \to -1.8 \to 4.92 \to -4.488$ runs away from 1.

**Pitfalls**

- Rounding $x_n$ to two or three places between steps. The rounding error then swamps what the method gains; keep six places or more.
- Stepping uphill. The update subtracts $\eta f'(x_n)$; adding it climbs away from the minimum.
- Taking $k$ to be the coefficient of $x^2$. It is $f''$: here 6, not 3, so the bound is $\eta < \frac{1}{3}$, not $\eta < \frac{2}{3}$.

**In the notes.** Unit 8: Newton's method, and Unit 8: Gradient descent.

**Practise it.** the Review set on this method